

Jimmy Energy Builds Scalable and Secure HPC for Nuclear Heating Simulations on AWS

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“Using AWS, we built something powerful while minimizing our effort.”

—Jeremy Picosson, Head of Scientific Simulation, Jimmy Energy



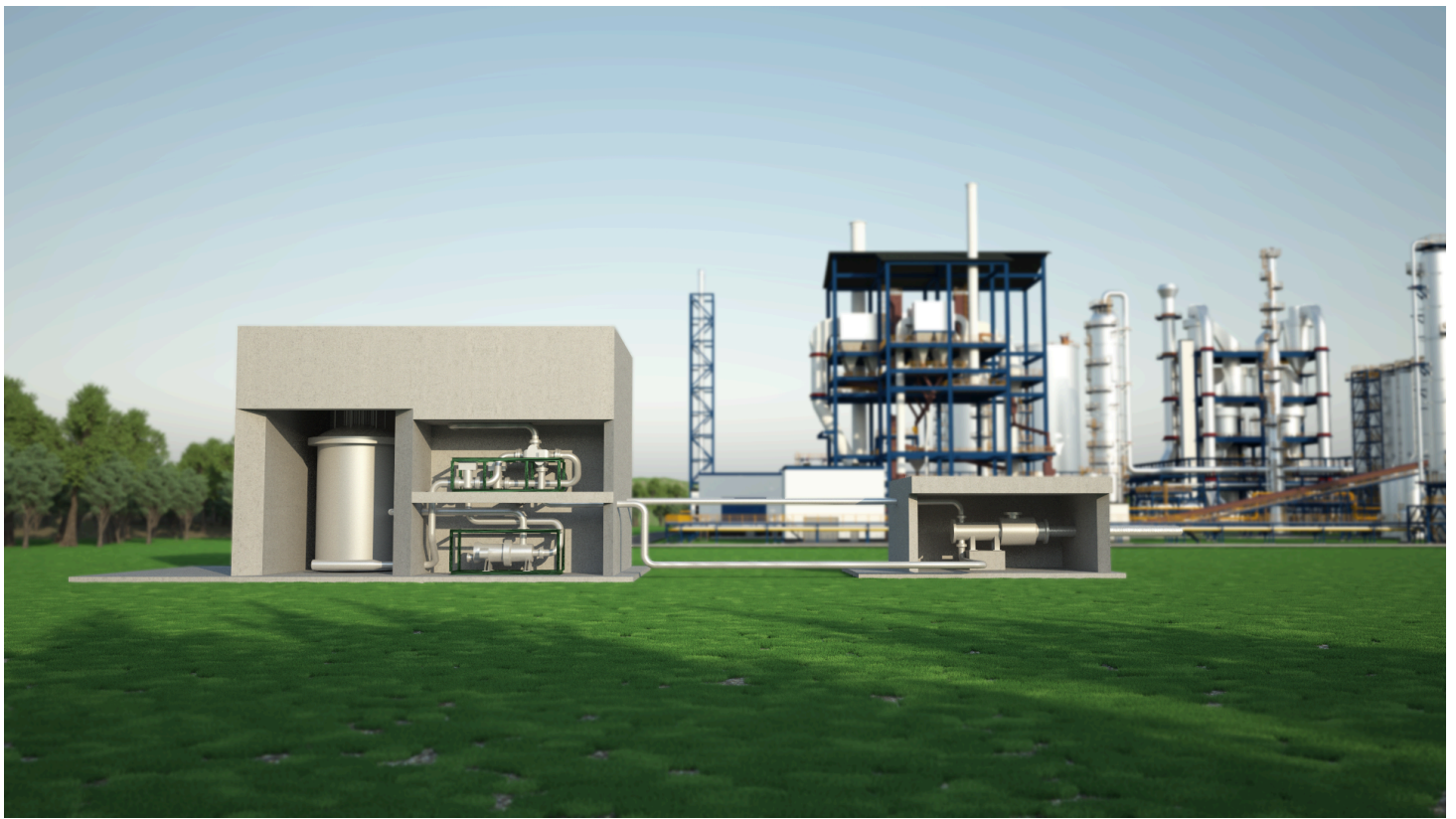
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[Jimmy Energy](#) (Jimmy) has been turning up the heat, with nuclear power taking center stage in its massive decarbonization campaign. Its small-scale generators can reduce carbon footprints across multiple industries, but to facilitate their safe and efficient operation, the company needed a reliable high performance computing (HPC) solution. Using Amazon Web Services (AWS), Jimmy implemented a natively built solution that optimized time savings while maintaining scalability and security.

Creating a Path for Industrial Decarbonization

Jimmy, a French startup created in 2020, set out to replace gas and coal burners with modular nuclear reactors that produce heat up to 550°C. Over the course of 20 years, a single reactor can save 300,000 tons of carbon dioxide emissions compared with traditional forms of energy.

To demonstrate the safety of its generators, Jimmy needs to establish that its reactors will operate safely and securely, even in an emergency. To do so, the company runs scientific simulations that require scalable and secure HPC. However, Jimmy couldn't build the infrastructure it needs for HPC on premises due to time and cost constraints. Jimmy chose AWS because it offered the most extensive variety of HPC services. "We were able to implement AWS services quite simply with limited resources," says Jeremy Picosson, head of scientific simulation at Jimmy. "Our HPC solution works well, and it would have required far more time and effort to build on premises."



Saving Time and Money Using AWS

One key element of Jimmy's HPC solution is [AWS ParallelCluster](#), an open-source cluster management tool used to deploy and manage HPC clusters on AWS. The company's simulations require up to 1.5 TB of memory and use 200 CPUs per calculation, so scalability was key.

Jimmy saved at least 3 months of development time and significant financial resources by using AWS for its HPC, which is vital for a startup where time to market is a priority. Over 1.5 years of using AWS to create the HPC cluster, Picosson only had to dedicate 10 percent of his time to building infrastructure. The simplicity of AWS helped Jimmy to focus instead on enhancing the safety and efficiency of its reactors. "Using AWS, we built something powerful while minimizing our effort," says Picosson. "We were able to build our infrastructure without having to wait months for it to work, and we can gradually improve it over time."

Jimmy established a secure solution for HPC by using AWS to encrypt its storage for data protection. "There are a lot of built-in solutions on AWS that make it simple to configure security," says Picosson.

Continuing to Improve Infrastructure on AWS

Now that the company has achieved its scalability goals with HPC, it can focus on furthering cost optimization and enhancing automation. "We knew AWS had the most capacity for HPC," says Picosson. "It was simple to implement and satisfied our needs."

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